

CCA GAME DEVELOPMENT · SESSION 4

# Modern Art Generator

Let's turn Python code into a living, colorful digital canvas!



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# What are we building today?



## 1. Memory Refresh

Quick check on PyGame Window coordinates **(0,0)** and RGB color mixing.



## 2. Randomness

Learn how to bring "surprise" into our code using the **random** tool.



## 3. For Loops

Learn how to tell the computer to repeat actions instantly.



## 4. Final Project

Build your own automated Modern Art Generator in VS Code!

# The Rules of Our Digital Canvas

## Screen Coordinates

In PyGame, the math is slightly different from school!

- **The Origin :** `(0, 0)` Always lives at the **Top-Left** corner.
- **X-Axis:** Moves **RIGHT** as the number gets bigger.
- **Y-Axis:** Moves **DOWN** as the number gets bigger.

## RGB Colors

We paint shapes by mixing light values.

- Format: `(Red, Green, Blue)`
- Values range from **0 (Dark)** to **255 (Bright)**.
- Example: `(0, 255, 0)` !



# Randomness

## Concept 1: Why Do We Need Randomness? 🎲

- **The Problem:** If games always look the same, players get bored! We need unpredictability.
- **The Solution:** We build a "dice roll" into our code.
- **The Tool:** Type `import random` at the top of your file.
- **The Command:** `random.randint(min,max)`

```
import random
#"Hey computer, pick a random number between 0 and 800!"
random_x = random.randint(0, 800)
random_y = random.randint(0, 500)
color = ( random.randint(0, 255), random.randint(0, 255), random.randint(0, 255))
```



# 🧩 Mini Practice 1: The Safe Border

## The Scenario:

Your game window is **800 X 600 pixels**. You want to spawn a box of size **50x50** randomly.

**Rule:** The box **MUST NOT** go off-screen!

## The Question:

If the box is 50px wide, what is the maximum safe limit for the X coordinate so it doesn't get cut off on the right side?

## Starter Code:

Fix the question marks in your editor!

```
import random
# TODO: Fix the max range so the 50px box stays inside the
800px window!

box_x = random.randint(0, ???)

print("Safe Box X:", box_x)
```

# For Loops

## Concept 2: The Power of For Loops

- **The Problem:** Typing `pygame.draw.rect()` 100 times manually is exhausting.
- **The Solution:** Hire a digital robot to repeat the command instantly!
- **The Command:** `for i in range (N)` tells Python how many times to repeat.

```
import random
#“Robot, repeat this drawing action 50 times in a split
second!”
for i in range(50):
    #drawing code goes here ...
    print("Drawing shape number:", i)
```



## Mini Practice 2: The Step Counter

### The Scenario:

You want a loop to run **10 times**. But instead of printing standard numbers (0 to 9), you want to print scaled steps: **0, 10, 20, 30...**

**Rule:** The box MUST NOT go off-screen!

### The Question:

How do you use the loop variable **i** to scale the output without changing **range(10)** ?

### Starter Code:

Use math to fix the print statement!

```
for i in range (10)

# TODO: Multiply 'i' by a number to scale the step!

print("Coordinate step:", ???)
```

## Mini Practice 3: The Fusion

Write a script using a **for loop** to draw **50 squares** on your PyGame window. Every square must have a random position and random color.

### The Rule:

Every square must have a **completely random position** (x,y) and a **random RGB color**.

### The Mission:

Write a script using a **for loop** to draw **50 squares** on your Pygame window.



### 1. Setup the Loop

Start with **for i in range(50)** to tell your computer to draw 50 items.



### 2. Randomize

Inside the loop, generate random X, Y positions and random **R, G, B** colors.



### 3. Draw!

Use **pygame.draw.rect()** using the random variables you just created.

# Mini Project 1: Modern Art Canvas



## The Requirements:

1. Keep your Pygame window open using the standard game loop (`running = True`).
2. Use a loop to generate at least **150+ random geometric shapes** (mix rects and circles!).
3. **Creative Touch:** Add a solid background color first before drawing your abstract art.

Run it in VS Code and watch your computer create unique art!



# What Did We Master Today? 🎉

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## Coordinates System

Pygame uses `(0,0)` at the top-left corner. Y goes DOWN, X goes RIGHT.



## Randomness

Using `random.randint()` brings surprise, variety, and unpredictability to our logic.



## For Loops

`for i in range()` automates repetitive tasks instantly, saving us from writing hundreds of lines of code.



## Next Session Preview...

Next week, we are going to learn how to make our shapes move across the screen!